

Apply for a Science Corps Fellowship - Entry #3303

Name

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Gender

Female

City, state, and country of residence

Oxford, England, United Kingdom

Nationality

India

When did or will you finish your PhD?

April 2025

University Education

I have been pursuing higher education in STEM, for more than a decade now. My educational experience is from elite universities such as University of Oxford, Imperial College London, Ludwig Maximilian University, University of Turin, and University of Hyderabad. All of this experience is summarized in the attached CV. Studying in different countries, like India, Italy, Germany and the United Kingdom, has provided me a great opportunity to appreciate different educational systems and styles. My keen observation has led me to evaluate the teaching practices that help students in better grasping of scientific concepts and remember for longer retention periods.

What are your reasons for applying to be a Science Corps fellow?

Growing up in India, with limited resources for science education, especially for practical application of science, studying science was never a tangible experience. India does have decent scientific resources, but these are mostly limited to elite institutions, which only 0.5% of extremely talented students

who can withstand gruesome competitive entrance exams get access to. For example, before I got into University of Hyderabad, I never had a scientific laboratory experience, leaving me question the real-life applicability of science. Even though I always had a predilection for sciences, I only started enjoying science after I started applying it to solve real problems. Hence, I strongly believe that to nurture young minds and to retain their curiosity for science, it is of utmost importance to employ teaching practices that encourage students to get hands on experience on handling scientific tools/equipment and to emphasize on practical demonstration of science, rather than just teaching theory and leaving the rest to student's imagination.

I deeply resonate with the mission of Science Corps and I am confident that it is an excellent platform, to use all of my experience and insights and to contribute to the development of science education for underprivileged communities.

In addition to your studies, describe any relevant experiences that have prepared you for a Science Corps fellowship.

I believe that an important part of knowledge acquisition, is also to give back and spread or pass on that knowledge to others. And there is no better way to do that than teaching others.

From a very young age of 14, I conducted mathematics, physics and chemistry tuitions to my neighbors in our community. When I grew older, during exams period at university, I gave crash courses that helped my friends to quickly revise the concepts learnt in classroom.

During my master's education, my classmates and seniors who joined the materials science course with a background in mechanical engineering struggled to cope up with quantum mechanics lectures. Me being from a physics background, I was able to clearly understand the concepts taught in the classroom. I offered my colleagues tuitions and retaught them quantum mechanics in my style which led them all to pass their exam with good scores.

During my PhD, I have supervised 3 master students 2 bachelor students and passed on the knowledge of my project to one post doc. Teaching about same project to students at different levels of their education, has helped me tailor my teaching practices to their level of expertise.

Where do you imagine yourself professionally in the future? Check all that apply.

Academic scientist in academia

Doing science in industry

In science education

In science outreach

Other

Expand on your selection above and describe where you imagine yourself professionally in the near (two year) and long-term (ten year) future.

Professionally, in near future, I imagine myself to be working for industry in research and development sector. This will enable me to apply the science I learnt over the years to develop cutting-edge technologies to solve global problems, such as energy crisis.

My long-term goal is to return to India, and to establish a school that provides high quality education with innovative teaching practices, that prepares the students for real world workplace challenges. For example, my school education was mostly focused on memorizing stuff. In today's world of internet, I believe it is not super important to memorize a formula, but rather to use that formula for a specific application. However, this application-oriented education is currently lacking in early school education in India. In the school that I envision, students would be trained to improve their analytical skills, critical thinking and logical reasoning. More emphasis would be on team building, communication, leadership, organizational and entrepreneurship skills. My goal is to design a school curriculum such that, students are inspired, and their curiosity is nurtured from a young age, so that they can excel in their future.

CV/Resume

 [CV-PA-August-2024.pdf](#)

How did you find out about Science Corps?

University email list

Any other comments/questions?

In the email I received about this position, I saw that one of your host locations is India. However, on your website, I could not find any information about this host site. So, I would imagine that it is your latest host location. I would like to know, when would this be functional. If provided an opportunity, I would like to serve at the host location in India.

[Science Corps](#)